technische universität dortmund

fachschaft statistik

Case Study II: Styrian Health Data

Week 3 Update: Modelling Improvements

Rohith B Chowdappa
Nixon M Paulson
Maria Cherian

Focus of This Week

- Continued from last week's EDA and initial Random Forest modelling
- Refined the model based on feedback from the previous presentation
- Focused on multi-output blood pressure prediction
- Added temporal and weather-based feature engineering
- Extended the work with classification and residual diagnostics

Feedback Addressed

- Moved away from relying only on the default train-test split
- Focused on the multi-output model instead of separate models only

Updated Modelling Pipeline

- Train-test split was stratified by age groups
- Random Forest depth was tuned to reduce overfitting
- Cross-validation was used to check model stability
- New temporal variables:
  - hour, weekday, month, season
- External weather variables:
  - temperature and humidity, matched by measurement hour

External Weather Data

- Hourly weather data was downloaded from the Open-Meteo Historical Weather API
- Location used: Graz, Austria
- Time period matched to the health measurements in 2006
- Variables added:
  - temperature at 2m
  - relative humidity at 2m
- Weather timestamps were matched to measurement timestamps rounded to the nearest hour

Source: Open-Meteo Historical Weather API

Model Robustness Checks

5-fold CV, baseline RF

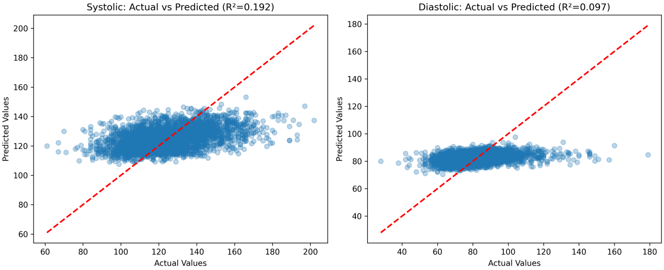
Target	Mean R^2	Std. dev.
Systolic	0.140	0.013
Diastolic	0.035	0.014

Tree depth experiment

Max depth	Mean R^2
3	0.138
5	0.157
10	0.163
20	0.140
None	0.140

- Moderate tree depth performed best
- Very deep trees showed signs of overfitting

Regression Performance Improvement



Temporal and environmental features improved predictive performance for both systolic and diastolic BP.

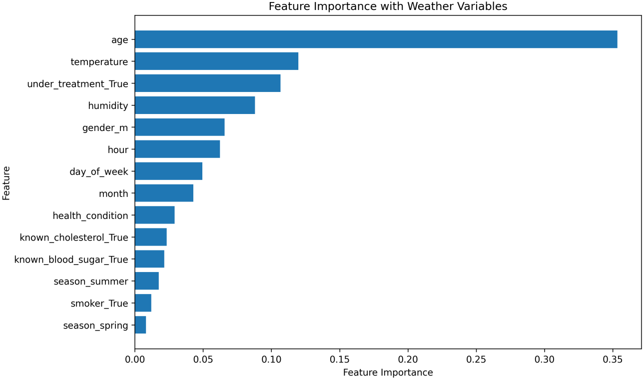
The improvement was especially noticeable for diastolic BP.

Prediction Error Comparison

Model	Sys MAE	Dia MAE
Original RF	14.1	10.5
Temporal RF	13.8	10.2
Weather RF	13.5	9.9

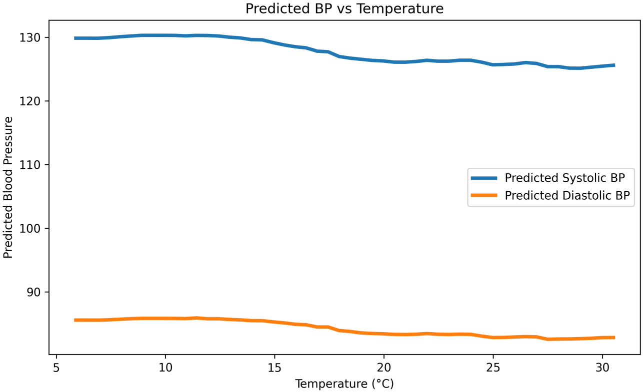
- MAE represents average prediction error in mmHg
- Prediction error decreased after temporal and weather integration
- Scatterplots remain noisy, but average prediction quality improved

Feature Importance with Weather Variables



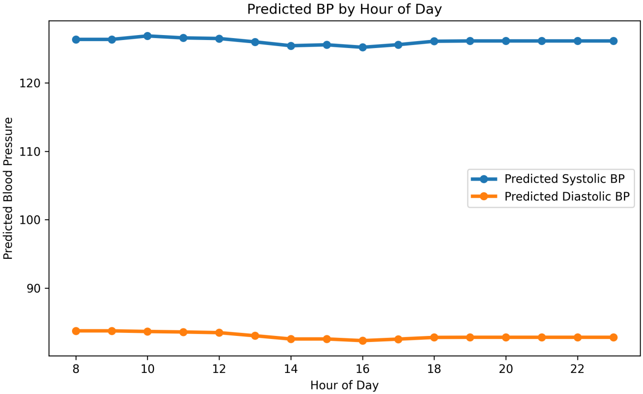
Age remained the strongest predictor, but temporal and environmental variables also contributed predictive information.

Predicted BP vs Temperature



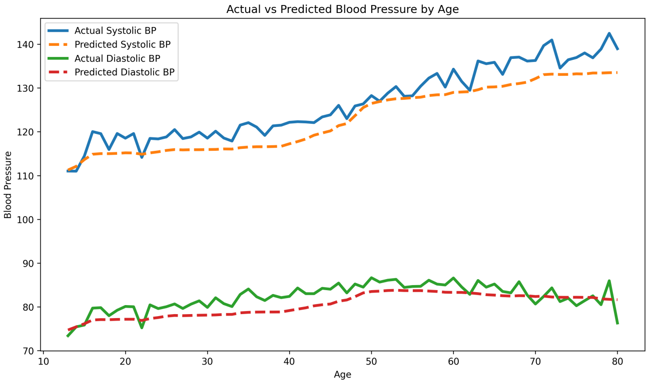
The weather-enhanced model learned a weak inverse association between temperature and predicted BP. This should be interpreted as predictive association, not causality.

Predicted BP by Hour of Day



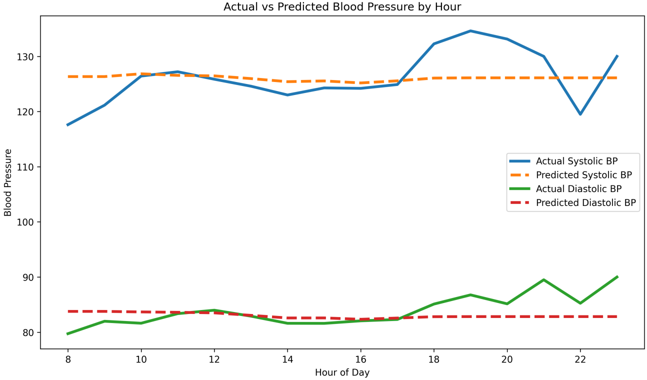
The model learned modest within-day variation. Hour alone does not dominate prediction, but it adds useful temporal context.

Actual vs Predicted BP by Age



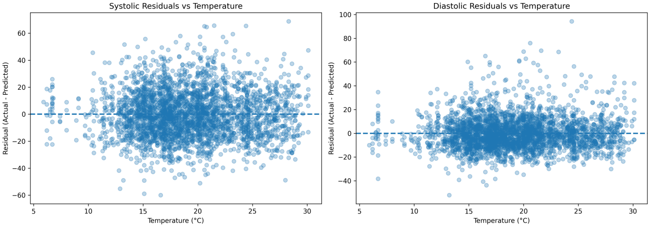
The model captures broader age-related trends, especially for systolic BP, while smoothing noisy individual variation.

Actual vs Predicted BP by Hour



The model captures the general hourly level, but sharp short-term fluctuations remain difficult to predict.

Residual Diagnostics



Residuals remain roughly centered across temperature values, suggesting no strong systematic temperature-related prediction bias.

Classification Extension

- Regression predicts exact BP values
- Classification predicts elevated BP categories
- Systolic high BP:

$$BP_{sys} \geq 140$$

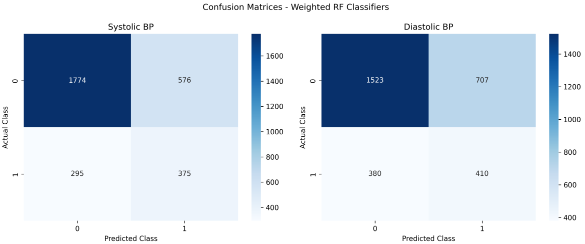
- Diastolic high BP:

$$BP_{dia} \geq 90$$

- Weighted Random Forest was used because high BP cases were less frequent

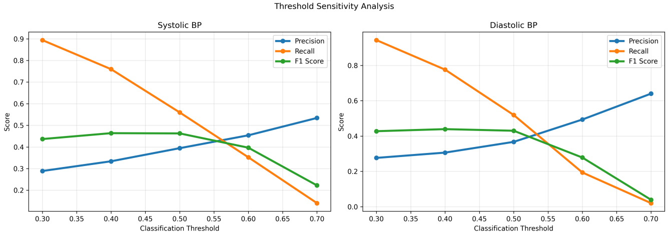
Classification Results

Target	Accuracy	Precision	Recall	F1
Systolic BP	0.712	0.394	0.560	0.463
Diastolic BP	0.640	0.367	0.519	0.430



Systolic classification performed slightly better, but both targets show a clear precision–recall tradeoff.

Threshold Sensitivity Analysis



Lower thresholds increase recall and detect more elevated BP cases, while higher thresholds reduce false positives.

Current Takeaways and Open Questions

- Temporal and weather features improved regression performance
- The model captures population-level trends better than individual values
- Classification gives a useful risk-category perspective
- Threshold choice strongly affects false positives and false negatives
- Open question: should final focus be regression performance, classification, or model interpretation?